

- Potassium sulphate [K_2SO_4]
- Calcium Ammonium Nitrate (CAN) also called as **Kisan Khad**.
- N, P and K are the macronutrients provided by inorganic fertilisers. Such fertilisers are also called as **complete fertiliser**.

2. Biofertilisers

The fertilisers, which are biologically originated are referred to as biofertilisers. In other words, biofertilisers are those organisms, which can bring about nutrient availability to crop plants. One of the important type of biofertiliser is manures.

Manures

- They are the semi-decayed substances, which are added to the soil to enhance soil's fertility.
- It is decomposed product of organic wastes.
- Manures improve the following properties of soil:
 - Fertility
 - Aeration
 - Water holding capacity
 - Crumb structure
- Farmers in our country are traditionally using cow dung as manure.
- They are also of different types based on their composition.

The three types of manures are as follows:

- Farm Yard Manure (FYM)** Consist of mixture of cattle dung, crop residues, fallen leaves, twigs, etc.
- Composite Manure** (compost) Consist of rotten vegetables, garbage, sewage sludge, animal refuse, etc.
- Green Manure** Quick growing crops cultivated and ploughed into the soil. These can increase next crop yield by 30-50%.

Green Manuring

Green manure is a type of cover crop grown primarily to add nutrients and organic matter to soil.

- The technique of utilising green manure is called **green manuring**, e.g. leguminous plants such as *Crotalaria juncea*, *Glycine max*, *Indigofera linifolia*, etc., are used as green manures.
- *Anabaena azollae* (cyanobacteria) is used as biofertiliser in rice field.
- Sunnhemp plants are used for green manuring in India.

Advantages of Manuring

- It improves the fertility of soil and enhances the soil structure.
- It improves the soil aeration capacity.
- It helps in control of pests, insects, mites and growth of weeds.

Sources of Biofertilisers

- Natural processes fix about 190×10^{12} g per year of nitrogen through **biological nitrogen-fixation**.
- Biological nitrogen-fixation provides about 1750 million tonnes of nitrogen, free of cost naturally in the form of biofertilisers.
- Biofertilisers contain some organisms, which can bring about nutrient availability to the crop plants.

The main sources of biofertilisers are:

1. Free-living Nitrogen-fixing Bacteria

Non-symbiotic (free-living) nitrogen-fixing bacteria are found abundantly in rhizosphere (i.e. the region where soil and roots make contact). They may be aerobic or anaerobic.

- Aerobic nitrogen-fixing bacteria** Aerobic nitrogen-fixing bacteria such as *Azotobacter* are thought to maintain reduced oxygen conditions (microaerobic conditions) through their high level of respiration. Others such as *Gloeotheca* evolve O_2 photosynthetically during the day and fix nitrogen during the night.
- Anaerobic nitrogen-fixing bacteria** In anaerobic nitrogen-fixing bacteria, oxygen does not pose a problem, because it is absent in their habitat. They can be either photosynthetic (e.g. *Rhodospirillum*) or non-photosynthetic (e.g. *Clostridium*).

2. Symbiotic Nitrogen-fixing Bacteria

Many plants form a symbiotic relationship with nitrogen-fixing bacteria that contain an enzyme complex nitrogenase. Nitrogenase enzyme can reduce nitrogen into ammonia.

- The best known example of symbiotic nitrogen-fixation is that of the *Rhizobium* group of bacteria, which form nodules on the roots of leguminous plants.
- Nodules contain an oxygen binding haemoprotein called **leghaemoglobin**.
- Leghaemoglobin is present in the cytoplasm of infected nodules cells at high concentration and gives the nodules pink colour.
- Non-leguminous trees and shrubs such as *Alder*, *Ceanothus* and *Purshia* have the nitrogen-fixing Actinomycetes, i.e. *Frankia* as a symbiont.

3. Free-living Nitrogen- fixing Cyanobacteria

Anabaena, *Nostoc*, *Aulosira*, *Cylindrospermum*, *Trichodesmium*, etc., are some free-living nitrogen-fixing blue-green algae or cyanobacteria.

- They add 20-30 kg of nitrogen per hectare of soil and water.
- *Aulosira* is the most active nitrogen fixer in rice field, whereas *Cylindrospermum* is active in sugarcane and maize fields.

4. Symbiotic Nitrogen- fixing Cyanobacteria

Anabaena and *Nostoc* sp. of cyanobacteria are common symbionts in lichens, *Anthoceros*, *Azolla* and *Cycads* roots.

- The water fern *Azolla* has *Anabaena azollae* as the symbiont, which fixes nitrogen within fronds.
- They grow thickly in rice paddy water and often inoculated to rice fields for nitrogen-fixation.

5. Mycorrhiza

Mycorrhizae (sing. mycorrhiza) are the mutualistic association of a fungus and the roots of higher plants. Mycorrhizae are found in most groups of vascular plants. There are two types of mycorrhizae listed as follow:

- (i) **Ectomycorrhiza** The mycorrhizal fungi is present on the root surface only. The cortical cells are surrounded by a network of fungal hyphae called **Hartig net**. It is found in roots of oak, *Pinus*, peach, *Eucalyptus*, etc.
- (ii) **Endomycorrhiza** The mycorrhizal fungus enters the roots and spreads in the root cortex. It may be intercellular or intracellular.

The type of endomycorrhiza, which produce vesicles and arbuscules inside the cortical cells of roots are called **VAM** (Vesicular Arbuscular Mycorrhiza), e.g. orchids, common crop plants and grasses.

Weed

Wild plants, which grow along with a cultivated crop are called **weeds**. Common weeds found in wheat and rice field are *Phalaris minar* (mandusi), *Amaranthus* (chaulai), *Chenopodium* (bathua). Exotic weeds are *Parthenium hysterophorus*, *Eichhornia*, etc. Most troublesome terrestrial weed is congress grass (*Parthenium hysterophorus*).

Weedicides

These are the chemical substances that are used to kill weeds. Some weedicides are butachloro 2,4-D (2,4-dichlorophenoxy acetic acid), **MCPA** (2-Methyl 4-Chloro, 1-Phenoxy Acetic acid).

Pesticides

Pesticides are used to destroy pests (pathogens, insects, rodents, etc.) by spraying the crop.

- Some microbes such as bacteria, fungi, viruses, protozoans, etc., and their products used for pest control are known as **biopesticides**.
- These are inexpensive, cause no pollution, pick no risk to human health and have a long term capability to control pests.
- Some common pesticides are warfarin sulphur, malathion, **DDT** (Dichloro Diphenyl Trichloroethane), BHC (Benzene Hexachloride), etc.
- Some examples of pesticides are:
 - DDT and 2, 4-D were mainly used during the Second World War.
 - Endosulfan is an organochlorine pesticide and acaricide used on plants and insects.
 - **Bordeaux mixture** a fungicide, composed of CuSO_4 , Ca(OH)_2 was discovered by **Millardet**.
 - **Methyl isocyanate gas**, which is responsible for Bhopal gas tragedy (3rd Dec, 1984), is used for synthesising carbaryl (selvin).

- Rotenone is obtained from roots of *Derris elliptica* and *Lonechocarpus*.
- Fumigation is another method of pest control used mainly to kill pests attacking stored grains. The substances used for fumigation are known as fumigants, e.g. methyl bromide, iodoform.

ANIMAL HUSBANDRY

Animal husbandry is the science of rearing, caring, feeding, breeding, improvement and utilisation of domesticated animals. It is an agricultural practice of breeding and raising **livestock**.

- The animals used for transport, milk, meat and agriculture are collectively called **livestock**.
- On the basis of utility, the livestock can be categorised into the following categories:
 1. **Milk yielding animals** Cows, buffaloes and goats provide us milk, which is used to obtain animal protein and serves as a perfect natural diet.
 2. **Meat and egg yielding animals** Sheeps, goats, pigs, chicken, ducks and fowls provide us meat and eggs.
 3. **Animals utilised as motive power** Buffaloes, horses, donkeys, bullocks, camels and elephants are used in transport and ploughing the fields.
 4. **Wool giving animals** Sheeps are reared for obtaining wool.
 5. **Miscellaneous uses** The hides of cattles are used for making a variety of leather goods.

Animal Breeding

- It is an important aspect of animal husbandry, which aims to increase the yield of animals and to improve the desirable qualities of produce.
- **Breed** is a group of animals related by descent and similar in most characters like general appearance features, size, configuration, etc.
- Animal breeding can be classified as:
 - (i) **Inbreeding** The crossing of closely related animals within the same breed for 4-6 generation is called **inbreeding**.
 - (ii) **Outbreeding** Breeding between unrelated animals either of the same breed but not having common ancestors for 4-6 generations (called **outcrossing**) or of different breeds (**cross-breeding**) or even different species (**interspecific hybridisation**) is called **outbreeding**.



ARTIFICIAL INSEMINATION

It is the process in which the stored semen of a superior male animal is introduced into the genital tract of a selected female animal by using suitable instruments, for carrying out reproduction, to get a better breed of the animal or to solve the problem of infertility.

Pisciculture

The breeding, rearing and transplantation of fish by artificial means is called pisciculture. It involves raising fish commercially in tanks or enclosures usually for food.

Fishes are of two types:

- (i) **Freshwater fishes** Rohu, *Catla*, carp, tiriya, mali, singhara and calbasu.
- (ii) **Marine fishes** Bombay duck, conger eel, pomphret, Salmon, Sardine, hilsa.

Fishes are the rich source of protein and vitamin (A and D).

Poultry

The term 'Poultry' means rearing of fowls, ducks, geese turkeys and some varieties of pigeons, but more often is used for fowl rearing.

Fowls are reared for food or for their eggs.

- (i) Poultry birds, reared for meat are called **broilers**.
- (ii) **Layers** are the female fowls raised for egg production.
- (iii) The fowls may be categorised into two breeds.
- **Indigenous breeds** The term 'Desi' includes all indigenous fowls exhibiting great variations in their shape, size and colour, e.g. Assel, Ahagus, etc.
- **Exotic breeds** The term 'Improved' is applied generally to the exotic and comparatively modern breeds of chickens, e.g. white leghorn, Rhode island red, etc.



SOME HIGH YIELDING BREEDS OF ANIMALS

- **Cows** Karan Swiss, Sunandini, Jersey-Sindhi, Brown Swiss- Sahiwal, Friesian-Sahiwal, etc.
- **Buffalo** Murrah, Surti, Jaffarabadi, Nili, Nagpuri, etc.
- **Goat** Kashmiri, Gaddi, Bengal, etc.
- **Sheep** Gaddi, Lohi, Bikaneri, Gurez, Nillore, etc.
- **Pig** Large white Yorkshire, Middle white Yorkshire, etc.
- Halstein is the cattle breed that produces highest amount of milk.
- Pashmina (fine wool) obtained from mountain (Kashmiri) goat.
- Indigenous Hens Aseel, Basara, Chittagong, Ghagus, Brahma and Cochin.
- Exotic high egg yielding breeds of hens White leghorn, Rhode island red, Black minorica, Plymouth rock and Light sussex.
- High yielding breeds of hens developed in India by cross breeding of indigenous varieties with the foreign breeds are ILS-82, HH-260 and B-77.

Beekeeping (Apiculture)

- It includes the maintenance of hives of honeybees for the production of honey and beeswax.
- **Honey** It can be used as a food of high nutritive value. A number of ayurvedic medicines contain honey.

- **Beeswax** Produced by honeybees is used in industry for the manufacture of cosmetics and polishes.
- Several species of honeybees are reported in different parts of India, but the most common species reared by beekeepers is *Apis indica*.

GENETIC ENGINEERING

Biochemist **Paul Berg** is the father of genetic engineering.

Genetic engineering or genetic manipulation involves the introduction of foreign DNA or synthetic genes into the organism of interest, so as to obtain desired product, e.g. Golden rice is a variety of *Oryza sativa* produced through genetic engineering to biosynthesise β -carotene, a precursor of provitamin-A in the edible parts of rice. Two β -carotene biosynthesis gene were transformed in golden rice.

- (i) ψ (*psy*) (phytoene synthase) from daffodil.
- (ii) *crt I* from soil bacterium *Erwinia uredovora*.

Later on, other modifications were also developed and golden rice was expected to reach market by the year 2013.

- **Polymerase Chain Reaction (PCR)** Discovered by **Kary Mullis** (1985). It is a technique of amplification of DNA segment with the help of *taq* polymerase (isolated from *Thermus aquaticus*).
- **Restriction enzymes** These are used to cut DNA at specific sites, also called **molecular scissors**.
- **Bioasphalt** It is an asphalt alternative made from non-petroleum based renewable resources like sugar, natural tree and gum resins, etc.
- **Gene therapy** It is the insertion of genes into an individual's cell or tissues to treat disease(s).
- **Terminator technology** It is a technology that is used to alter seeds, which fail to germinate after the first generation.
- **ELISA** It is a test for AIDS, which is based on Ag-Ab reaction.

Some Achievements of Genetic Engineering

- **Sweet bug** A bacterium that turns coconut water into delicious confectionary. The bacterium is *Acetobacter oxylinum*.
- **Prof Anand Mohan Chakraborty** has developed a new strain of oil eating bacteria called 'Superbug' by using species of *Pseudomonas*.
- **Organic foods** Foods produced by using methods free of inputs such as synthetic pesticides, chemical fertilisers, chemical additives, irradiation, etc.
- **Biodegradable plastic films (PSV)** made by blending the polymers with an additive to provide an oxidation or starch. First human insulin **Humulin** was produced by genetic engineering.

DNA Fingerprinting

It is a technique employed by forensic scientists to assist in the identification of individuals by their respective DNA profiles. This technique is based upon the fact that the DNA constitution of an individual carries some specific sequences of nucleotides, which do not carry any information for protein synthesis. These are called VNTR regions, i.e. Variable Number Tandem Repeats or mini satellites.

DNA fingerprinting was invented by **Dr Alec Jeffreys**.

Process

- Collection of sample (most preferably buccal swab).
- **RFLP** (Restriction Fragments Length Polymorphism) analysis.
- **Southern blotting** DNA fingerprinting is a quite laborious method that requires large amount of undegraded sample so the method was replaced by what is used today based on STR (Short Tandem Repeats) and PCR.

Cloning

Cloning in biology is the process of producing similar organism of genetically identical individuals that occur in nature when organisms such as bacteria, insects or plants reproduce asexually.

There are two types of cloning:

- Reproductive cloning** It is used for the production of offspring. In it a new individual is created from a single cell by replacing the nucleus in an egg cell with the nucleus (containing the genetic material) from another cell of the body. The cloned egg cell grows and develops into an embryo. The embryo is implanted inside a surrogate mother's womb to mature and produce a viable foetus.
- Therapeutic cloning** It is used for the development of embryonic stem cells. The inner cell mass of the blastocyst is then removed and used for the creation of an embryonic stem cell line that has the genetic make up of the donated nucleus.

SUCCESSFULLY DEVELOPED CLONED

- Dolly a sheep, the first mammal clone synthesised by Dr Wilmut, UK.
- **Noori** The world's first cloned Pashmina goat named as Noori, an Arabic word referring to light. Funded by World Bank, the clone project was jointly worked by SKAUST and Karnal based National Dairy Research Institute (NDRI).
- The world's first buffalo calf through the Handguided Cloning Technique was born on 6 February, 2009 at NDRI, Karnal.
- **Garima-II**, another cloned buffalo calf through the new and advanced 'Handguided Cloning Technique' was born at NDRI, Karnal.

Stem Cell Therapy

The most promising use of stem cells is due to their ability to be modified into different functional adult cell types and serve as a potential source of replacement cells to treat numerous diseases. Stem cell research involves the use of several different types of cells besides embryonic stem cell, such as adult stem cells from humans or animals or stem cells from foetuses, umbilical cord or amniotic fluid.

Transgenic or GM Crops

A transgenic crop is a crop, which contains and expresses a transgene (a gene that is transferred into an organism by genetic engineering). It involves introduction of new DNA into a crop to confer a new superior trait. A popular term for the transgenic crop is genetically modified crops or GM crops.

Bt Crops Bt crops are produced by inserting the toxic genes that code for a toxin directly into the genetic code of the crop in order to protect the crop from insects along with this, section of gene known as a **promoter**, which would encourage the crop to produce the toxin and a **marker**, which could be used to track and identify modified crop are also introduced, e.g. *Bt* cotton, *Bt* brinjal, Golden rice, etc.

Bt Cotton It is created by using a bacterium, *Bacillus thuringiensis* (*Bt* is short form). It produces a protein that kills certain insects, e.g. beetles, flies, budworm, etc.

Golden Rice It is a transgenic rice biosynthesising β -carotene, vitamin-A, essential for eye sight precursor in the edible parts of rice.

Advantages of GM Crops

- Provides high nutrient containing varieties, i.e. enriched in specific nutrients.
- GM crops are more tolerant to cold, insect, drought, salinity, etc.
- Resistance to pest, herbicides and diseases.

Hybridisation

It is a technique, which involves cross breeding of different varieties of crop to get a new crop with desired characteristics. Such crops/varieties are called **High Yielding Varieties** (HYV) or hybrid.

- High yielding varieties of **wheat** are Lerma, Sharbati Sonara, Kalyan Sona, Roja 64, Pusa Lerma, Hera-Moti, Sonara-64, Sonalika and Arjun.
- High yielding varieties of **paddy** (rice) are Jaya, Padma, Pusa 205, IR-6, T-141.
- High yielding varieties of **maize** are Ganga-101, Ranjit, Daccan hybrid.
- **Germ plasm** It is the living genetic resource such as seed or tissue maintained for the purpose of animal and plant breeding preservation and other research aspects.

Terms Related with Biotechnology

- **Genetic screening** is an experimental technique used to identify genes that may produce damaging changes in the individuals.
- It involves collecting sequencing and comparison of DNA genomes.
- **Biochips** DNA chips single-stranded DNA chains or repetitive DNA segments fixed on a silica glass chip.
- **Bioinformatics** An interdisciplinary field, which addresses biological problems using computational techniques and makes the rapid organisation and analysis of biological data possible.
- **Blue biotechnology** An application of biotechnology in the marine and aquatic, but rarely used.
- **Green biotechnology** It is an application of biotechnology in the agricultural processes.
- **Red biotechnology** It is an application of biotechnology in the medical processes.
- **White biotechnology** It is a branch of biotechnology applied to industrial processes.
- **Biomarkers** (Biosignature) are specific proteins used to diagnose disease.

Green Revolution (1940-70)

Green revolution refers to a series of research, development and technology transfer initiatives that increased agricultural production around the world.

- Great increase in the production of food grain crops due to use of hybrid varieties, fertilisers and modern agricultural techniques took place.
- Father of green revolution in India is **MS Swaminathan** and in world is **NE Borlaug**.

Operation Flood (White Revolution)

Great increase in the production of milk due to use of high milk yielding breeds of cow and buffaloes.

- It resulted in making India the largest producer of milk and milk products and hence called **white revolution**.
- Operation flood was rural programme started by India's National Dairy Development Board (NDDB) in 1970.

Silver Revolution

Great increase in the production of eggs due to use of exotic breeds.

Fermentation

Fermentation is the conversion of carbohydrates to alcohol and carbon dioxide or organic acids using yeasts, bacteria or a combination of them, under anaerobic conditions, e.g.

Saccharomyces cerevisiae used in baking and alcohol industry.

- Cheese is formed by the process of coagulation of casein by enzyme rennin.
- Yoghurt is produced by fermentation using *Lactobacillus* and *Streptococcus*.

BIOGAS

Biogas is also called **Landfill Gas (LFG)** or **digester gas**.

It is the gas produced by biological breakdown of organic matter in the absence of O_2 .

- It is methane rich and produced by anaerobic digestion of biomass with the help of methanogenic bacteria.
- It is mainly produced from animal wastes.
- **Non-biodegradable waste** cannot be broken down by microorganism, so cannot be used for biogas production.
- **Organic wastes** such as dead plants or animal material, animal dung, and kitchen waste are also called **biodegradable waste** can be easily converted into gaseous fuel, i.e. biogas. These wastes include cow dung, paddy-husk, vegetables/fruits, wastes, etc.
- It mainly comprises methane CH_4 (50-70%) and carbon dioxide CO_2 (25-50%) and hydrogen sulphide H_2S (0-3%), moisture and siloxanes.

Advantages of Biogas

- It is cheaper than other fuels and ideal for small scale application.
- It helps to clean environment because waste materials can be used as substrate.
- It is a renewable source of energy.

BIOREMEDIATION

It is the treatment of environmental problems by the use of biological methods. It may be defined as treatment, which includes microorganisms or their enzymes to return the environment to its original condition. It may be employed in order to attack specific contaminants, such as chlorinated pesticides that are degraded by bacteria.

PHYTOREMEDIATION

This is the treatment of environmental problems through the use of plants.

- The term phytoremediation is a combination of two words, i.e. **phyto** means plant and **remediation** means to remedy.
- It helps in depolluting contaminated soils, water and air with plants, able to degrade or eliminate metals, pesticides, solvents, explosives, crude oil and its derivatives.